

sol2_implementer_deep_agents

The working filled Lab 2 loop. Take-home issue 118.

Python holds Pass / Retry / Escalate. Deep Agents is the maker. The red gate is `junit.xml`.

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What this folder is

FILE	ROLE
<code>harness.py</code>	<code>red_gate</code> , <code>run_loop</code> , CLI
<code>implementer.py</code>	eight-step loop
<code>doers.py</code>	<code>none</code> / <code>reference</code> / <code>cli</code> / Deep Agents
<code>rubric.py</code> / <code>gates.py</code> / <code>contract.py</code>	copies, not a library
<code>roles.py</code>	<code>create_deep_agent</code>
<code>tests/</code>	judge read-only, refuse <code>tests/**</code>

Why it exists

Saturday fills three stubs. There is no drop-in `harness.py`. This folder is the shipped eight-step answer.

`task loop:implementer` is gone from the root Taskfile. Demo from here. Each criterion becomes two nodes in `steps.jsonl`.

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Learning objectives

- Walk `implementer.run` step by step
- Prove `--doer none` escalates
- Prove `--doer reference` can pass under WriteScope
- Configure Deep Agents so the judge holds `read_file` only
- Grep for `from loops` and fail if it returns

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Starting architecture

```
orchestrator  writes nothing, owns budget
planner       steps.jsonl
test_implementer  tests/**
code_implementer  app/**, denied tests/**
judge          read_file only
```

↓

red gate → ten-row rubric → gates.decide

↓

.harness/last-implementer.json

Eight steps in `implementer.run`

1. Read ticket. Refuse if not `ready`.
2. `plan_for`: one test step and one code step per criterion. Derived, not generated. This is Graph Engineering. Not LangGraph.
3. `test_implementer` writes under `tests/**`.
4. Red gate. Empty `new-ids` → escalate.
5. `code_implementer` writes `app/**` until green.
6. Ten-row rubric. No model.
7. `judge_done=None`, so a green rubric is enough. Session 2 still teaches a model judge.
8. `gates.decide`. Trace to `.harness/last-implementer.json`.

Red gate

```
def _new_test_ids(before: set[str], after_failed: set[str]) -> set[str]:  
    return {test_id for test_id in after_failed if test_id not in before}
```

A test that already existed and still fails is not proof of a new contract.

`--doer none` hits this every time. If that run were green, the harness would be lying.

Deep Agents wiring

```
return create_deep_agent(  
    model="anthropic:claude-sonnet-5",  
    tools=[run_tests_tool(repo)],  
    subagents=subagents_for(contract, loop),  
)
```

Judge tools = ["read_file"]. Code-implementer write tool refuses tests/** with a sentence.

create_deep_agent **does not** count retries. Python still owns gates.decide.

Commands

```
cd solutions/sol2_implementer_deep_agents
python3 -m pytest tests -q
python3 harness.py --table-only
python3 harness.py --repo ../../work/northwind-field-crm --ticket T001 --doer none
python3 harness.py --repo ../../work/northwind-field-crm --ticket T001 --doer reference
python3 harness.py --repo ../../work/northwind-field-crm --ticket T001 --doer deep
```

--doer deep **needs** deepagents and a key. Tests do not.

Expected results

--doer none:

```
gate: escalate  
reason: red gate: no new test was observed failing.
```

--doer reference: **copies** `known-good` **into** `tests/**` **then** `app/**`, each phase bound by that role's WriteScope. Ten PASS rows. gate: `pass`.

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Troubleshooting

SYMPTOM	CAUSE	FIX
<code>task loop:implementer</code> missing	engine deleted	run this <code>harness.py</code>
<code>--doer none</code> is green	red gate not stopping	empty new-ids must escalate
Wrote <code>tests/**</code> as coder	write tool not scoped	sentence refusal
<code>from loops</code> in a file	design leaked	<code>test_standalone.py</code> fails the build

Validation

- `[] pytest:judge tools == ["read_file"]`

- `[] coder write to tests/test_due.py returns REFUSED`

- `[] _new_test_ids({"old"}, {"old","new"}) == {"new"}`

- `[] same signature twice → ESCALATE`

- `[] --table-only prints judge with no repo`

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Recap

Same role table as Saturday. Enforcement is a missing tool, plus a path check, plus Python on the outside.

If a port imports `loops`, the design leaked.

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